Heliyon 9 (2023) e14514

Contents lists available at ScienceDirect

Heliyon

journal homepage: www.cell.com/heliyon

Research article

Effect of addition of organic manures on basmati yield, nutrient content and soil fertility status in north-western India

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ARTICLE INFO

Keywords:
Basmati rice
Organic manures
Soil fertility
Fertilization
Nutrient availability

ABSTRACT

Integrated use of chemical fertilizers with organic manure is becoming a quite promising practice to maintain higher crop productivity and to manage soil health, which is otherwise deteriorated due to intensive cultivation and imbalanced fertilizer use. Thus, the present experiment was conducted for combined application of inorganic fertilizers and organic manures for higher yield and nutrient uptake in basmati rice as well as to restore soil health. The treatments applied in the present study involve T1: control, T2: Farmyard manure (15 t ha^{-1}), T3: Poultry manure (6 t ha^{-1}), T4: Press mud (15 t ha^{-1}), T5: Rice straw compost (6 t ha^{-1}), T6: Farmyard manure (15 t ha^{-1}) + 50% N (recommended dose of nitrogen), T7: Poultry manure (6 t ha^{-1}) + 50% N, T8: Press mud (15 t ha^{-1}) + 50% N, T9: Rice straw compost (6 t ha^{-1}) + 50% N, T10: 75% N, T11: Farmyard manure (15 t ha^{1}) + 75% N, T12: Poultry manure (6 t ha^{-1}) + 75% N, T13: Press mud (15 t ha^{-1}) + 75% N, T14: Rice straw compost (6 t ha^{-1}) + 75% N, T15: 100% N. The integrated use of organic manures and inorganic fertilizers significantly increased the grain and straw yield, macronutrients as well as micronutrients uptake in basmati due to the release of nutrients from decomposition of organic manures. Among different treatments, the addition of PM+75% N showed maximum grain yield, straw yield and sustainability yield index (44.53 q ha^{-1}, 89.67 q ha^{-1} and 0.91 respectively) as well as the highest uptake of nitrogen (58.29 and 65.39 kg ha^{-1}), phosphorus (25.04 and 23.24 kg ha^{-1}) and potassium (15.26 and 118.95 kg ha^{-1}) in grain and straw, respectively. Similar results were observed for zinc (Zn), copper (Cu), iron (Fe) and manganese (Mn) uptake under treatment involving PM+75% N with values 183.9, 26.18, 339.3 and 355.8 g ha^{-1} in grain and 205.3, 25.62, 2627.3 and 278 g ha^{-1} in straw, respectively. Additionally, correlation studies showed that the grain and straw yield of basmati exhibited a significantly positive correlation with soil P, Cu and Fe. The study suggested that the partial substitution of inorganic fertilizers with organic manures did not lower crop yield and nutrient uptake. Thus, integrated application of organic and inorganic fertilizers can be used for the sustainability of basmati-wheat system and to retain the soil fertility which is otherwise deteriorated with sole use of inorganic fertilizers.

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https://doi.org/10.1016/j.heliyon.2023.e14514
Received 3 July 2022; Received in revised form 5 March 2023; Accepted 9 March 2023
Available online 15 March 2023
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1. Introduction

Rice (*Oryza sativa* L.)-wheat (*Triticum aestivum* L.) cropping system (RWCS) is the most technologically advanced extensive system which provides staple foods to the world's population [1–3]. In Asia, RWCS is grown in 13.5 million hectares (mha) of land, with 57% of it in South Asia [4]. In India, RWCS has maximum area in Indo-Gangetic Plains (IGPs) which contributes major portion in the national food grain production. The area under aromatic rice is about 2.1 M ha land in India [5]. The RWCS is associated with the deterioration of natural resources, declining factor productivity, nutrients deficiencies, groundwater depletion, and agricultural sustainability. Additionally, the long term and imbalanced use of nutrients has caused nutrient mining from the soil, deteriorated crop productivity and ultimately the soil health. To overcome the declining productivity, many farmers have already resorted to use excessive doses of inorganic fertilizers which are resulting in the deterioration of soil health [6]. Therefore, adoption of the alternate nutrient management practices may prove beneficial to a greater extent in the current situation for yield sustainability of RWCS and improving the soil health. Combined use of organic and inorganic fertilizers is useful practice to continue production going forward without compromising the system's yield [7,8].

The application of organic manures improves the soil fertility by supplying essential plant nutrients, especially micronutrients, which boosts the crop growth [9,10]. Manures can substitute mineral fertilizer for increasing crop productivity, carbon sequestration, soil structure, as well as reducing environmental pollution [11,12]. Previously, it was well-reported that manure fertilization significantly improved the physicochemical and biological properties of the soil, such as pH, bulk density, enzymatic activity, soil aggregation, soil organic carbon, and both macro- and micro-nutrients [13]. Manure fertilization can improve the soil's physical structure, allowing it to store more water and nutrients, and thus increase crop productivity [14,15] and have residual effect on succeeding crop [16,17]. Moreover, partial substitution of organic manures with inorganic fertilizers will lower the production cost to a greater extent [18]. Previous studies have reported that substitution of chemical fertilizers with organic manure have beneficial effects on crop yield [19,20]. Use of different manures (press mud, poultry manure and farmyard manure) in rice-wheat system has evidenced to increase the crop yield and DTPA-extractable micronutrients (Zn, Cu, Fe and Mn) concentrations in soil of north-western India [21,22]. In long-term field experiment of 40 years under soybean-maize rotation, the application of organic manures on brown soil improved the crop yields along with soil properties [23].

Apart from organic manures, residue retention has also been promoted as a viable option for soil amendment to increase the crop yield, while reducing the use of chemical fertilizers. With the increase in crop production, the amount of crop residue produced per year has also been increased. Returning crop residue to the soil helps to improve the organic matter content, soil physico-chemical and biological properties, thus sustain soil fertility. Moreover, the retention of crop residue can increase crop yields and quality. The existing literature has shown positive impact of crop residue incorporation on crop yield. For instance, addition of rice residue on *Typic Ustochrept* sandy loam soil of north-western India has significantly increased the yield of wheat crop as compared to the untreated plots, with simultaneous improvement in soil quality [24]. Another study reported the beneficial effect of maize residue incorporation on the seed yield of soybean [25]. Under the rice-wheat cropping system, the retention of crop residue has increased the micronutrient content in the crop [21].

To the best of our knowledge, the information on different paired comparisons of fertilization involving organic and chemical fertilizers under basmati-wheat cropping system on north-western India is still lacking. The specific objective of the present study was to determine the most effective paired comparison of organic manures and inorganic N fertilizer to improve yield and nutrient (N, P, K, Zn, Cu, Fe and Mn) uptake in basmati rice. The treatments involved the different pairs of organic manures (FYM, poultry manure (PM), pressmud)/crop residue with chemical fertilizers. The present study may prove beneficial as the partial substitution of chemical fertilizers with organic manures in basmati rice will improve the crop yield and micronutrient concentration and also improves the soil nutrient status.

Table 1
Treatment details for application of organic manures and fertilizers to basmati.

Treatment	Manure	Fertilizer
T1	-	-
T2	**FYM (15 t ha^{-1})**	-
T3	**PM (6 t ha^{-1})**	-
T4	**PrM (15 t ha^{-1})**	-
T5	**RSC (6 t ha^{-1})**	-
T6	**FYM (15 t ha^{-1})**	**50% RDN**
T7	**PM (6 t ha^{-1})**	**50% RDN**
T8	**PrM (15 t ha^{-1})**	**50% RDN**
T9	**RSC (6 t ha^{-1})**	**50% RDN**
T10	-	**75% RDN**
T11	**FYM (15 t ha^{-1})**	**75% RDN**
T12	**PM (6 t ha^{-1})**	**75% RDN**
T13	**PrM (15 t ha^{-1})**	**75% RDN**
T14	**RSC (6 t ha^{-1})**	**75% RDN**
T15	-	**100% RDN**

FYM: Farmyard manure; PM: Poultry manure; PrM: Press mud; RSC: Rice straw compost; RDN: Recommended dose of nitrogen.

2. Materials and methods

2.1. Site specifications

The present study was conducted during 2019–20, 2020–21 at Research farm of Department of Soil Science, Punjab Agricultural University, Ludhiana (30° 56′ N, 75° 52′ E and 247 m above mean sea level) to analyse the effects of manures and fertilizers on basmati–wheat system. The region exhibited subtropical climate along with hot, rainy summers as well as dry winters. Annually the rainfall ranges between 400 nm and 600mm where July to September months receives maximum rainfall of around 70% [26]. The experiment comprised of 15 treatments in randomized block design with three replications (Plot size = 6.0 × 1.2 m^2). Different paired comparisons of fertilization including inorganic fertilizers and organic manures were applied (Table 1) using 40 kg ha^{-1} nitrogen (N) as 100% recommended dose of N (RDN). The organic manures were added to the soil before transplanting of basmati. The nutrient content of different organic manures is given in Table 2. Basmati variety 'Punjab Basmati 5' was transplanted during the second week of July. The management practices and fertilizers were applied according to the package of practices [27]. After reaching the physiological maturity level, the plants were manually harvested, where the grain as well as straw samples were collected for examination. In the present experiment, the chemical characteristics of sandy loam soil such as pH, EC, OC, N, P, K, Zn, Cu, Fe and Mn contents were evaluated. The standard methods were followed to determine pH, EC and OC and were found to be 6.80, 0.16 dSm^{-1} and 0.19%, respectively [28–30]. The method by Subbiah and Asija, the Olsen extractable P method, and the method by Merwin and Peech were used to estimate the availability of N, P, and K to be 153 kg ha^{-1}, 23.9 kg $ha^{-1}$and 138 kg ha^{-1} in soil, respectively [31–33]. Also, the initial micronutrient (Zn, Cu, Fe, and Mn) levels in soil were 0.46 mg kg^{-1}, 0.28 mg kg^{-1}, 1.42 mg kg^{-1} and 4.19 mg kg^{-1}, respectively [34], as calculated using the instrument AAS (Varian AAS-FS 240 model).

2.2. Estimation of yield, macro and micronutrients in plant samples

Ten plants samples were tagged randomly from central rows to measure the plant height, panicle length, number of productive tillers per hill and 1000 grain weight of basmati. Plants were harvested manually at maturity and grain and straw yield were recorded. The sustainability yield index (SYI) was calculated using equation (1).

$$SYI = \frac{y - \sigma}{y_{max}} \tag{1}$$

Where y is average yield of treatment over two years, σ is standard deviation and y_{max} is observed maximum yield over two years.

The samples of grain and straw were oven dried for 3 days at 65 °C and grounded. The N content of samples was estimated using Kjeldahl's method [28]. For estimation of other nutrients macro (P, K) and micronutrients (Zn, Cu, Fe and Mn) concentrations, wet-acid digestion of dried grain and straw samples was done with diacid mixture of HNO_3 and $HClO_4$ in the ratio of 4:1 [35]. The total P concentration was estimated spectrophotometrically using vanado molybdo phosphoric yellow color method [36], total K using a flamephotometer [37] and micronutrients content using atomic absorption spectrophotometry method [38]. The concentration of different macro and micronutrients in manures were also analyzed using similar methods. The uptake of different nutrients was calculated using the formula given below (Equation (2)).

$$\text{Uptake } \left(\text{g ha}^{-1}\right) = \frac{\text{Concentration } \left(\text{mg kg}^{-1}\right) \times \text{Yield}\left(\text{q ha}^{-1}\right)}{10} \tag{2}$$

2.3. Nutrient use efficiency indices

The calculation for mobilization efficiency index (MEI) involved equation (3) as follows:

$$\text{MEI} = \frac{\text{Nutrient cocnentartion in grain } \left(\text{mg kg}^{-1}\right)}{\text{Nutrient concentration in straw } \left(\text{mg kg}^{-1}\right)} \tag{3}$$

The determination of physiological efficiency of N was done through equation (4) given below [26].

$$\text{PE} = \frac{\text{Y}_\text{t} - \text{Y}_\text{c}}{\text{NU}_\text{t} - \text{NU}_\text{c}} \tag{4}$$

Table 2
Nutrient content in organic manures used in present study.

Organic Manure	N (%)	P (%)	K (%)	Zn (mg kg^{-1} DM)	Cu (mg kg^{-1} DM)	Fe (mg kg^{-1} DM)	Mn (mg kg^{-1} DM)
FYM	1.91	0.740	2.09	115.7	63.9	650.1	149.2
PM	3.20	1.81	1.65	135.2	56.9	520.8	432.8
PrM	2.19	0.79	0.64	144.2	95.3	486.1	294.1
RSC	1.70	0.41	1.59	134.8	52.1	520.7	203.4

FYM: Farmyard manure; PM: Poultry manure; PrM: Press mud; RSC: Rice straw compost; DM: Dry matter.

where, Y_t and Y_c denotes grain yield (kg ha^{-1}) in N treated plots as well as in control, respectively; NU_t and NU_c denotes the total nutrient N uptake (kg ha^{-1}) in N treated plots as well as in control, respectively.

2.4. Statistical analysis

The experimental data was subjected to statistical analysis by using SPSS version 16.0 (SPSS Inc., Chicago, USA) packages. Analysis of variance (ANOVA) and Duncan's multiple range test was used to compare the least significant difference (LSD).

3. Results

3.1. Effect of fertilizers enriched manures on plant growth and yield attributes

The effect of sole and combined use of manures and fertilizers on plant growth and yield attributes of basmati has been presented in Table 3. The field investigation data demonstrated that the plant height, panicle length, number of productive tillers per hill and 1000 grain weight values of basmati ranged from 118 to 130.5 cm, 20.3–25.7 cm, 19–22 and 21.3–29.5 g, respectively. The maximum plant height (130.5 cm) was observed in the treatment involving application of poultry manure+75% N (T12) which was significantly higher than control (118 cm). Thus, the application of poultry manure+75% N significantly increased the plant height, whereas other treatments were statistically identical with control. Likewise, panicle length was maximum in treatment T12 (25.7 cm) followed by T15 (25 cm) and minimum in treatment T1 (20.3 cm). Thus, the supply of nutrients either through poultry manure+75% N or 100% N significantly increased the panicle length, whereas other treatments showed non-significant variation as compared to untreated plot. In context to the number of productive tillers per hill, the effect of manures and fertilizers application was non-significant. The data further indicated that in terms of 1000 grain weight, each treatment performed significantly against control except treatment T6. Treatment T12 resulted in maximum 1000 grain weight (29.5 g) which was significantly higher than control, but statistically significant with all the other treatments.

3.2. Effect of fertilizers enriched manures on basmati yield and SYI

The mean of two-year data demonstrated that the average grain yield of basmati varied from 26.57 to 44.53 q ha^{-1} and straw yield varied from 55.09 to 89.67 q ha^{-1} (Table 4). For grain yield, the results of treatment involving application of poultry manure+75% N (T12) were statistically non-significant with treatment involving poultry manure + 50% N (T7), farmyard manure + 75% N (T11) and pressmud + 75% N (T13), whereas for straw yield the treatment T12 was statistically non-significant with treatment T8. The incorporation of organic manures significantly enhanced the grain and straw yield of basmati over the control. Also, the treatment including the application of poultry manure (T3, T7 and T12) recorded highest grain and straw yield over the other treatments with similar rate of N application. With the increase in rate of N fertilizer application, grain and straw yield of basmati enhanced significantly and thus the treatments in which 75% N were applied along with manures were found to be more proficient in enhancing the grain and straw yield. Also, nutrient supply through 100% N significantly enhanced the grain yield by 43.1% over the control. Sustainability refers to the maintenance of productivity on a long-term basis and used as a quantitative measure to assess sustainability of an agricultural

Table 3
Effect of different paired comparisons of manures and fertilizers on plant growth and yield attributes of basmati.

Treatments	Plant height (cm)	Panicle length (cm)	No. Of productive tillers per hill	1000 grain weight (g)
T1	118.0^{b}	20.3^{b}	19	21.3^{b}
T2	124.0^{b}	21.6^{b}	19	26.8^{a}
T3	127.4^{ab}	23.9^{ab}	22	28.6^{a}
T4	126.6^{ab}	22.9^{ab}	20	29.3^{a}
T5	126.0^{ab}	23.0^{ab}	20	27.2^{a}
T6	128^{ab}	22.8^{ab}	19	25.3^{ab}
T7	129.5^{ab}	24.6^{ab}	21	28.2^{a}
T8	126.3^{ab}	24.6^{ab}	19	27.9^{a}
T9	128.2^{ab}	23.6^{ab}	20	28.3^{a}
T10	127.3^{ab}	24.6^{ab}	20	26.5^{a}
T11	129.3^{ab}	24.6^{ab}	21.5	26.5^{a}
T12	130.5^{a}	25.7^{a}	21.5	29.5^{a}
T13	128.0^{ab}	24.8^{ab}	20.5	28.2^{a}
T14	127.5^{ab}	24.3^{ab}	20	28.8^{a}
T15	127.0^{ab}	25.0^{a}	20	27.3^{a}
LSD (P = 0.05)	6.2	3.2	NS	4.6

In column, mean with a similar or dissimilar letter(s) was evaluated with the least significant difference (LSD) multiple range tests using a probability level of $p \leq 0.05$. T1 (control), T2: Farmyard manure (15 t ha^{-1}), T3: Poultry manure (6 t ha^{-1}), T4: Press mud (15 t ha^{-1}), T5: Rice straw compost (6 t ha^{-1}), T6: Farmyard manure (15 t ha^{-1}) + 50% N (recommended dose of nitrogen), T7: Poultry manure (6 t ha^{-1}) + 50% N, T8: Press mud (15 t ha^{-1}) + 50% N, T9: Rice straw compost (6 t ha^{-1}) + 50% N, T10: 75% N, T11: Farmyard manure (15 t ha^{1}) + 75% N, T12: Poultry manure (6 t ha^{-1}) + 75% N, T13: Press mud (15 t ha^{-1}) + 75% N, T14: Rice straw compost (6 t ha^{-1}) + 75% N, T15: 100% N.

Table 4
Effect of different paired comparisons of manures and fertilizers on grain and straw yield of basmati.

Treatments	Grain yield (q ha^{-1})			Straw yield (q ha^{-1})			SYI
	I Year	II Year	Mean	I Year	II Year	Mean	
T1	25.14^{g}	28.00^{f}	26.57^{g}	55.30^{f}	54.88^{j}	55.09^{g}	0.55
T2	31.86^{f}	31.50^{de}	31.68^{f}	58.38^{f}	61.16^{i}	59.77^{g}	0.71
T3	40.24^{bde}	32.35^{c}	36.29^{d}	87.30^{a}	67.93^{g}	77.61^{d}	0.69
T4	37.66^{e}	29.50^{ef}	33.58^{ef}	81.14^{cd}	64.89^{h}	73.02^{e}	0.62
T5	31.11^{f}	29.75^{e}	30.43^{f}	73.01^{e}	68.43^{fgh}	70.72^{f}	0.66
T6	32.76^{f}	34.00^{c}	33.38^{e}	72.51^{e}	63.69^{hi}	68.10^{f}	0.73
T7	47.00^{a}	35.50^{bc}	41.25^{a}	76.71^{de}	70.22^{e}	73.47^{df}	0.74
T8	40.88^{bde}	34.25^{cd}	37.56^{cd}	86.73^{ac}	82.21^{abc}	84.47^{ab}	0.74
T9	41.63^{ad}	35.00^{c}	38.31^{bc}	83.64^{bcd}	73.25^{eg}	78.45^{bde}	0.76
T10	38.43^{cd}	35.50^{bc}	36.96^{de}	79.26^{d}	74.54^{def}	76.90^{cde}	0.78
T11	44.10^{ab}	39.00^{a}	41.55^{ab}	82.70^{bc}	81.51^{ac}	82.11^{bc}	0.85
T12	47.31^{a}	41.75^{a}	44.53^{a}	91.66^{a}	87.68^{a}	89.67^{a}	0.91
T13	42.16^{bc}	39.75^{a}	40.96^{ac}	82.70^{bc}	76.82^{bce}	79.76^{bd}	0.88
T14	40.14^{be}	39.00^{a}	39.57^{bd}	79.73^{d}	82.18^{ac}	80.95^{bc}	0.87
T15	37.34^{e}	38.70^{ab}	38.02^{cd}	87.76^{ab}	81.27^{acd}	84.52^{ab}	0.83
LSD (P = 0.05)	4.15	3.46	3.72	5.6	6.8	6.4	

In column, mean with a similar or dissimilar letter (s) was evaluated with the least significant difference (LSD) multiple range tests using a probability level of $p \leq 0.05$. T1 (control), T2: Farmyard manure (15 t ha^{-1}), T3: Poultry manure (6 t ha^{-1}), T4: Press mud (15 t ha^{-1}), T5: Rice straw compost (6 t ha^{-1}), T6: Farmyard manure (15 t ha^{-1}) + 50% N (recommended dose of nitrogen, phosphorus and potassium), T7: Poultry manure (6 t ha^{-1}) + 50% N, T8: Press mud (15 t ha^{-1}) + 50% N, T9: Rice straw compost (6 t ha^{-1}) + 50% N, T10: 75% N, T11: Farmyard manure (15 t ha^{1}) + 75% N, T12: Poultry manure (6 t ha^{-1}) + 75% N, T13: Press mud (15 t ha^{-1}) + 75% N, T14: Rice straw compost (6 t ha^{-1}) + 75% N, T15: 100% N.

practice. In the present study, the maximum SYI was recorded in treatment T12 (0.91) as given in Table 4. The minimum SYI value was observed for T1 (0.55) which indicated that the incorporation of manures and fertilizer either alone or in paired forms enhanced the sustainability of basmati rice.

3.3. Effect of fertilizers enriched manures on uptake of macronutrients

The results concerning the impact of addition of manures and fertilizers on the uptake of macronutrients in grain and straw are given in Fig. 1a and b, respectively. The data revealed that the macronutrient uptake (N, P and K) in grain and straw of basmati improved to a significant extent with the sole and combined application of organic manures and inorganic fertilizers. The uptake of N, P and K in grain ranged from 28.16 to 60.94 kg ha^{-1}, 12.24 to 25.04 and 7.22–15.26 kg ha^{-1}, respectively, under different treatments. However, in straw the N, P and K uptake ranged from 30.86 to 65.39, 9.72 to 23.24, and 61.37–118.95%, respectively. The macronutrient uptake in grain and straw were recorded highest in treatment T12, which was significantly higher than other treatments and control. Similarly, the highest values for P and K were observed under the treatment T12 involving the combined application of poultry manure+75% N with the uptakes of 25.04 kg ha^{-1} and 15.26 kg ha^{-1} in grain along with uptakes of 23.24 kg ha^{-1} and 118.95 kg ha^{-1} in straw of basmati, respectively. Also, the results of Fig. 3 demonstrated that MEI-N was maximum in treatment T7 (0.561) exhibiting the Poultry manure (6 t ha^{-1}) + 50% N application and was lowest in treatment T5 (0.430). Additionally, the results of PE-N were highest in treatment T2 with values 1.456 q g^{-1}, whereas, the lowest PE value was observed in treatment T12 (0.838 q g^{-1}). The results of MEI indicated that MEI of N was higher in the presence of poultry manure treatment along N application.

3.4. Effect of fertilizers enriched manures on uptake of micronutrients

The data pertaining to the uptake of micronutrients in grain and straw as affected by manures and fertilizers is given in Fig. 2a and b, respectively. The results demonstrated that the micronutrient uptake (Zn, Cu, Fe and Mn) in grain and straw of basmati increased significantly with the sole and combined application of organic manures and inorganic fertilizers over the control. The uptake of Zn, Cu, Fe and Mn in grain ranged from 90.3 to 183.9, 15.38 to 26.18, 135.8 to 339.3 and 135.5–335.8 g ha^{-1}, respectively, under different treatments. However, in straw the Zn, Cu, Fe and Mn uptake ranged from 92.6 to 205.3, 12.39 to 25.62, 1158.0 to 2627.3 and 134.4–402.3 g ha^{-1}, respectively. The Zn uptake in grain and straw were recorded highest in treatment T12 involving the application of poultry manure+75% N, which was significantly higher than other treatments and control. Similar trend was observed for Fe with uptake of 339.3 and 2627.3 g ha^{-1} in grain and straw of basmati, respectively. In case of Cu, maximum Cu uptake in grain and straw was also recorded in treatment T12 with values 26.18 and 25.62 g ha^{-1}, respectively, however the results were statistically identical with treatments T8, T11, T13, T14 and T15.

3.5. Effect of fertilizers enriched manures on soil properties and nutrient build-up

The impact of different paired comparisons of organic and inorganic amendments on soil properties are given in Table 5. The addition of organic manures and fertilizers did not significantly affect the soil pH. In case of EC, the addition of organic manures and

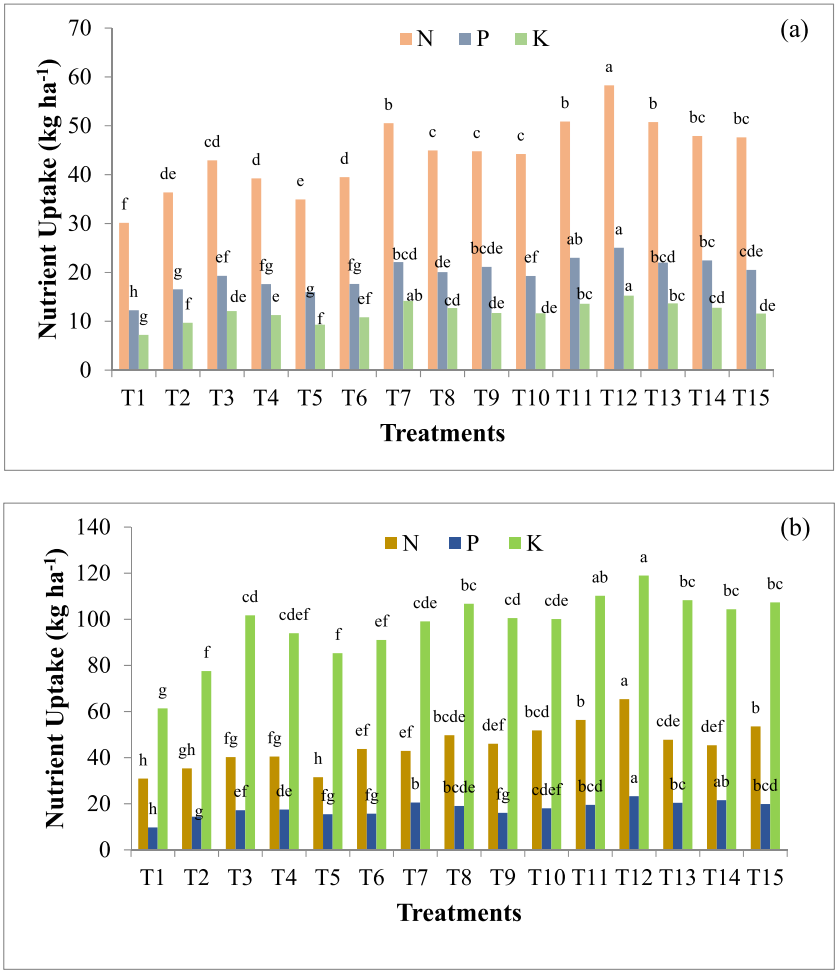


Fig. 1. Effect of different paired comparisons of manures and fertilizers on N, P and K uptake by (a) grain and (b) straw of basmati. T1 (control), T2: Farmyard manure (15 t ha^{-1}), T3: Poultry manure (6 t ha^{-1}), T4: Press mud (15 t ha^{-1}), T5: Rice straw compost (6 t ha^{-1}), T6: Farmyard manure (15 t ha^{-1}) + 50% N (recommended dose of nitrogen, phosphorus and potassium), T7: Poultry manure (6 t ha^{-1}) + 50% N, T8: Press mud (15 t ha^{-1}) + 50% N, T9: Rice straw compost (6 t ha^{-1}) + 50% N, T10: 75% N, T11: Farmyard manure (15 t ha^{1}) + 75% N, T12: Poultry manure (6 t ha^{-1}) + 75% N, T13: Press mud (15 t ha^{-1}) + 75% N, T14: Rice straw compost (6 t ha^{-1}) + 75% N, T15: 100% N.

inorganic fertilizers significantly resulted in increased EC values as compared to untreated soil. The highest EC was observed in T11 (0.39 dS m^{-1}), which was statistically at par with T15. The maximum value of OC was observed in treatment T6 (0.52%), which was statistically at par with treatments T8, T11 and T13 and was significantly higher than untreated soil (T1). The macronutrient data of soil demonstrated that the available N content in soil ranged from 152 to 228 kg ha^{-1} with the integrated application of fertilizers along with manures. The treatments with the addition of FYM and fertilizers showed the highest increase in N as compared to control and other treatment pairs. The available P content of the soil ranged from 27.67 kg ha^{-1} (T1) to 49.08 (T13) kg ha^{-1}. In the PrM +75% N treatment. Likewise, the available K content in soil ranged significantly from 79.4 to 143.0 kg ha^{-1}. The application of different manures either alone or in pair with chemical fertilizers resulted in a higher build-up of available K in soil as compared to the control.

The DTPA-extractable Zn in different treatments varied significantly from 0.86 to 1.89 mg kg^{-1} among different treatments. The maximum DTPA-extractable Zn was reported in the treatment T6 i.e. 50% N + FYM, which was statistically at par with 75% N + PrM and the minimum was recorded in T1. The results of T1 were statistically at par with T0 and T15 which showed that addition of inorganic fertilizers did not affect the Zn content in soil. The DTPA-extractable Cu ranged significantly from 0.53 to 0.93 mg kg^{-1} in different treatments. The highest DTPA-extractable Cu content was reported in the PM + 75% N treatment, which was significantly higher than all other treatments. The data pertaining to DTPA-extractable Fe varied from 19.36 to 30.52 mg kg^{-1} in all the treatments. The maximum DTPA-extractable Fe resulted from the treatment T12 which was statistically at par with T7, T8, T9, T11 and T14. The DTPA-extractable Mn content ranged from 2.21 (T1) to 3.86 (T14) mg kg^{-1} in different treatments. The results of T14 for DTPA-extractable Mn were statistically at par with the treatment T12 involving 75% N + PM.

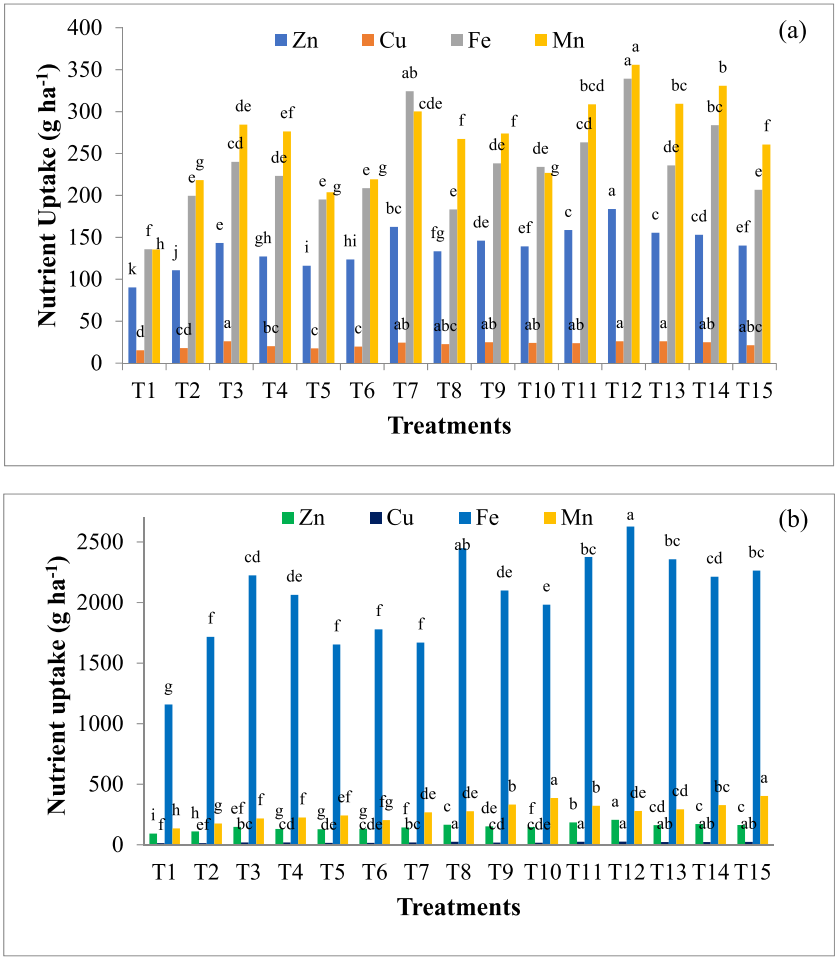


Fig. 2. Effect of different paired comparisons of manures and fertilizers on Zn, Cu, Fe and Mn uptake by (a) grain and (b) straw of basmati. T1 (control), T2: Farmyard manure (15 t ha^{-1}), T3: Poultry manure (6 t ha^{-1}), T4: Press mud (15 t ha^{-1}), T5: Rice straw compost (6 t ha^{-1}), T6: Farmyard manure (15 t ha^{-1}) + 50% N (recommended dose of nitrogen), T7: Poultry manure (6 t ha^{-1}) + 50% N, T8: Press mud (15 t ha^{-1}) + 50% N, T9: Rice straw compost (6 t ha^{-1}) + 50% N, T10: 75% N, T11: Farmyard manure (15 t ha^{1}) + 75% N, T12: Poultry manure (6 t ha^{-1}) + 75% N, T13: Press mud (15 t ha^{-1}) + 75% N, T14: Rice straw compost (6 t ha^{-1}) + 75% N, T15: 100% N.

3.6. Correlation among different soil properties and with yield

The correlation coefficients among different soil properties are given in Table 6. The results demonstrated that there was a significant positive correlation of OC with N (0.960), K (0.675) and Zn (0.549). The data given in Table 7 also demonstrates that the grain and straw yield of basmati was positively and significantly correlated with soil P (r = 0.600 and 0.614, respectively), Cu (r = 0.623 and 0.652, respectively) and Fe (r = 0.607 and 0.620, respectively). Thus, the present study showed that a strong correlation exists between yield and soil properties. Thus, alteration in soil properties due to the addition of manures and fertilizers improved the root growth, and thus increased yield due to supplementation of essential nutrients.

4. Discussion

4.1. Effect of fertilizers enriched manures on plant growth and yield attributes

The application of organic manures showed a positive impact on plant height and panicle length, however significant increase in case of poultry manure+75% N was observed over the control. The results for plant height might be due to the presence of high N content in manures, particularly in poultry manure, which favoured the vegetative growth of basmati [21]. The highest panicle length of basmati with application of poultry manure+75% N was possibly associated with the higher P content in poultry manure, whereas untreated plants showed least panicle length as compared to the treated ones. Phosphorus possesses high affinity to improve root anatomy that favours higher panicle formation development. The present results are in concordant with the previous studies [39]. As panicle length indicates the number of grains the plant can hold, thus increase in panicle length is associated with the increase in crop

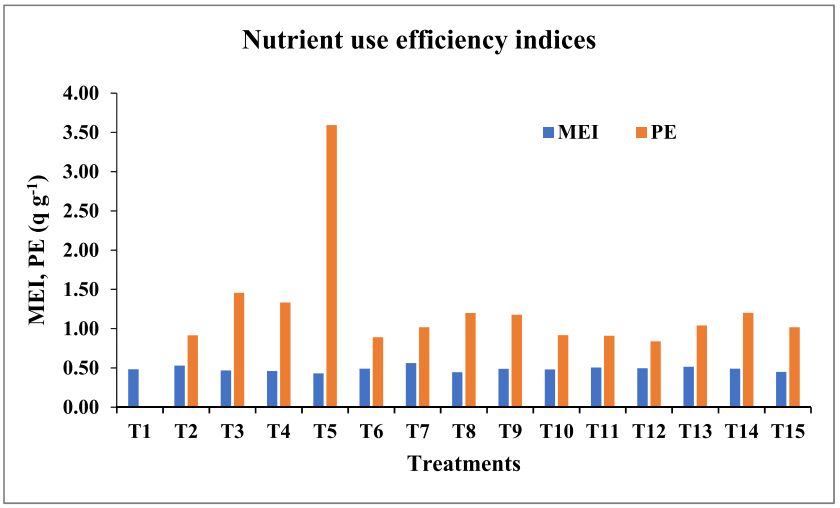


Fig. 3. Effect of different paired comparisons of manures and fertilizers on nutrient use efficiency indices. T1 (control), T2: Farmyard manure (15 t ha^{-1}), T3: Poultry manure (6 t ha^{-1}), T4: Press mud (15 t ha^{-1}), T5: Rice straw compost (6 t ha^{-1}), T6: Farmyard manure (15 t ha^{-1}) + 50% N (recommended dose of nitrogen), T7: Poultry manure (6 t ha^{-1}) + 50% N, T8: Press mud (15 t ha^{-1}) + 50% N, T9: Rice straw compost (6 t ha^{-1}) + 50% N, T10: 75% N, T11: Farmyard manure (15 t ha^{1}) + 75% N, T12: Poultry manure (6 t ha^{-1}) + 75% N, T13: Press mud (15 t ha^{-1}) + 75% N, T14: Rice straw compost (6 t ha^{-1}) + 75% N, T15: 100% N.

Table 5
Interactive effect of manures and fertilizers on soil properties.

Treat ments	pH	EC (dS m^{-1})	OC (%)	N (kg ha^{-1})	P (kg ha^{-1})	K (kg ha^{-1})	Zn (mg kg^{-1})	Cu (mg kg^{-1})	Fe (mg kg^{-1})	Mn (mg kg^{-1})
T1	7.68	0.27^{f}	0.35^{f}	152^{g}	27.67^{i}	79.4^{g}	0.86^{f}	0.53^{h}	19.36^{f}	2.21^{g}
T2	7.80	0.31^{de}	0.43^{cd}	190^{cd}	34.87^{h}	140.6^{a}	1.70^{bc}	0.69^{fg}	21.91^{ef}	2.76^{ef}
T3	7.95	0.31^{de}	0.39^{def}	169^{ef}	36.08^{h}	100.5^{e}	1.26^{d}	0.71^{efg}	20.64^{ef}	2.57^{f}
T4	7.75	0.29^{ef}	0.46^{b}	201^{bc}	41.87^{ef}	97.6^{f}	1.69^{bc}	0.79^{ce}	26.06^{b}	3.16^{cd}
T5	7.64	0.29^{ef}	0.38^{ef}	166^{f}	44.00^{cdef}	106.4^{def}	1.67^{bc}	0.88^{ab}	23.34^{cef}	2.97^{de}
T6	7.53	0.28^{ef}	0.52^{a}	228^{a}	40.90^{fg}	114.7^{cd}	1.89^{a}	0.81^{bcd}	26.22^{bc}	2.95^{de}
T7	7.98	0.33^{c}	0.41^{de}	179^{d}	45.78^{abcd}	119.3^{bc}	1.57^{c}	0.80^{bcd}	27.18^{a}	2.71^{f}
T8	7.64	0.36^{abc}	0.49^{ab}	216^{ab}	47.43^{abc}	107.8^{de}	1.64^{b}	0.79^{cde}	28.44^{a}	2.92^{de}
T9	7.65	0.27^{f}	0.42^{cde}	184^{de}	34.73^{h}	103.2^{ef}	1.00^{e}	0.82^{bcd}	27.41^{a}	3.47^{b}
T10	7.68	0.30^{ef}	0.38^{ef}	166^{f}	45.42^{bcde}	118.0^{b}	0.85^{f}	0.68^{g}	22.32^{def}	3.05^{de}
T11	7.53	0.39^{a}	0.51^{a}	225^{a}	48.10^{ab}	143.0^{a}	1.31^{d}	0.82^{bcd}	29.80^{a}	3.03^{de}
T12	7.78	0.33^{c}	0.46^{bc}	201^{bc}	42.94^{def}	103.9^{ef}	1.38^{d}	0.93^{a}	30.52^{a}	3.57^{ab}
T13	7.62	0.35^{bc}	0.49^{ab}	216^{ab}	49.08^{a}	126.3^{b}	1.77^{ab}	0.85^{abc}	25.57^{bcd}	2.61^{f}
T14	7.59	0.31^{de}	0.44^{c}	193^{c}	38.09^{gh}	102.7^{ef}	1.62^{b}	0.77^{cdef}	29.08^{ab}	3.86^{a}
T15	7.60	0.37^{ab}	0.39^{def}	171^{ef}	42.00^{ef}	101.3^{ef}	0.92^{f}	0.75^{defg}	22.17^{def}	3.41^{bc}
LSD (P = 0.05)	NS	0.03	0.04	15	3.58	9.8	0.17	0.08	3.49	0.30

In column, mean with a similar or dissimilar letter (s) was evaluated with the least significant difference (LSD) multiple range tests using a probability level of $p \leq 0.05$. T1 (control), T2: Farmyard manure (15 t ha-1), T3: Poultry manure (6 t ha-1), T4: Press mud (15 t ha-1), T5: Rice straw compost (6 t ha-1), T6: Farmyard manure (15 t ha-1) + 50% N (recommended dose of nitrogen, phosphorus and potassium), T7: Poultry manure (6 t ha-1) + 50% N, T8: Press mud (15t ha-1) + 50% N, T9: Rice straw compost (6 t ha-1) + 50% N, T10: 75% N, T11: Farmyard manure (15t ha1) + 75% N, T12: Poultry manure (6 t ha-1) + 75% N, T13: Press mud (15 t ha-1) + 75% N, T14: Rice straw compost (6 t ha-1) + 75% N, T15: 100% N.

yield. Thus, the nutrient composition of organic manures was responsible for the observed trend. For 1000 grain weight, the maximum values with incorporation of poultry manure+75% N was probably due to the enhanced water holding capacity and nutrients availability that enhanced the photosynthesis and transportation of carbohydrates to grains. Thus, it eventually increased the grain-size and thus 1000 grain weight. The present results are in line with the results obtained in literature [21], where addition of manures along with inorganic fertilizers increased the plant growth parameters. Thus, the present results demonstrated that for optimum growth of plant, 100% N can be replaced with some portion of manures.

4.2. Effect of fertilizers enriched manures on grain and straw yield

The increase in grain and straw yield of basmati with the combined application of manures and fertilizers might be due to improved soil fertility. The exogenous supply of organic matter through manures improved soil health status by supplying macronutrients and

Table 6
Correlation between yield, macronutrient and micronutrients uptake in basmati with nutrient concentration in soil.

	pH	EC	P	K	OC	Zn	Fe	Mn	Cu	N
EC	−0.137									
P	−0.174	.646**								
K	−0.085	0.494	.515*							
OC	−0.13	0.442	0.434	.675**						
Zn	−0.002	0.004	0.329	0.343	.549*					
Fe	0.069	0.308	.623*	0.133	0.433	.563*				
Mn	−0.326	0.142	0.283	−0.039	0	0.112	0.36			
Cu	−0.104	0.281	.637*	0.274	0.467	0.473	.858**	.550*		
N	−0.151	0.447	0.445	.676**	0.960**	.551*	0.437	0.005	0.473	

Table 7
Correlation of grain and straw yield of basmati with different soil properties.

	pH	EC	OC	N	P	K	Zn	Cu	Fe	Mn
Grain	0.058	.613*	0.207	0.213	.600*	0.345	0.012	.623*	.607*	0.366
Straw	−0.131	.598*	0.166	0.178	.614*	0.078	−0.086	.652**	.620*	.561*

increasing microbial activity which releases different organic acids that help in solubilisation of native soil nutrients and make available for the uptake by plants. Besides this, the secondary and micronutrients were also added to soil which were not otherwise supplied by inorganic fertilizers [40]. Thus, combined use of organic manures and chemical fertilizers might have enhanced the bioavailability of nutrients in soil as well as altered soil physical properties which resulted in improved root growth and water absorption capacity [41]. Additionally, quick release of N from poultry manure as compared to the other organic manures can be attributed to its superior performance as compared to the other organic manures. Nitrogen influenced the grain yield through a source-sink relationship and in addition to higher production of photosynthates, it resulted in increased translocation to reproductive parts. Nitrogen being a most important plant nutrient needed for growth and development of plant was known to increase the yield of basmati [42]. Thus, the use of organic manure with chemical fertilizers has proved beneficial for regulating the soil fertility and gradual release of plant essential nutrients for maintaining productivity of basmati under basmati-wheat system [43,44].

4.3. Effect of fertilizers enriched manures on uptake of macronutrients and NUE

The increase in macronutrients uptake might be due to the increased yield of grain and straw as well as the introduction of certain amount of N through organic manures. The different results of N uptake under different treatments were due to the different nitrogen content in organic fertilizers as well as their different mineralization rates in the soil [45]. Poultry manure enhances soil characteristics, allowing for the establishment of a larger root density and reduced leaching, which enhances the crop's nutrient-absorption capability thus increasing nutrient uptake in crop. The N uptake by plants also increased due to its involvement in various metabolic activities which are required for plant growth and produce more dry matter [46]. Nitrogen plays a major role in the synthesis of chlorophyll and amino acids, which are the important building blocks of protein synthesis. Similar results were observed for the uptake of P and K due to higher grain yield and nutrient content. The addition of organic fertilizers increased the organic matter content of soil, which on decomposition released the macronutrient in soil and thus increased their availability. Another work reported that integration of organic fertilizer with reduced dose of recommended N significantly increased the *N*-uptake in rice [47]. The higher N uptake by rice might be due to the higher yield received in these treatments involving organic manures and inorganic fertilizers [48]. The results of present study were in concordant with the previous studies where application of organic manures and fertilizers increased the uptake of macronutrients in plant [49,50].

4.4. Effect of fertilizers enriched manures on uptake of micronutrients

The higher uptake of micro-nutrients with application of organic manures and inorganic fertilizers might be attributed to the higher bioavailability of micronutrients in soil, which increased their uptake in basmati as compared to control [51]. The breakdown of organic matter releases nutrients through various mechanisms, which increases their availability in soil [52]. Moreover, addition of organic manures leads to lowering in soil pH due to the release of various acids, which favours the enhanced bioavailability of micronutrients as compared to the sole use of inorganic fertilizers. Similar results have been reported in which higher uptake of micronutrients have been recorded with combined use of inorganic and organic fertilizer treated plots as compared to control and sole use of inorganic fertilizers [8,53,54]. Also, synergistic application of both organic and inorganic fertilizer may serve a sufficient level of nutrition which takes part in biosynthesis of chlorophyll. Chlorophyll biosynthesis is mainly affected by light, temperature, carbohydrate and availability of N. In the present study, the addition of farmyard manure and N nutrients increased the growth as well as production of plant because these materials increased the growth of roots, reinforced the plant stems, and enhanced the photosynthetic rates thus increasing the amount as well as uptake of micronutrients by crop.

4.5. Effect of fertilizers enriched manures on soil properties and nutrient build-up

The addition of organic manures increased soil EC due to the increase in soluble salt concentration sourced from feed additives. The decomposition of manures releases acids which chemically react with sparingly soluble salts and increase solubility. The increase in OC was due to the addition of organic matter through manures. The results recorded a significant increase in NPK content with organic manures and fertilizers. The lower macronutrient status in control treatment was attributed to the mining of available macronutrients by crops without fertilization for 2 years. The improved macronutrient levels in manure added plots could probably be attributed to the nutrient supplemented to the soil from the decomposition of organic matter and releases CO_2. In case of P, the organic acids released during the decomposition of organic manures, accelerate the P solubilisation due to enhanced microbial activities, and due to retarded P fixation in soil. The higher K content in the soil might be associated with the increased cation exchange capacity of soil which improved the capacity of soil to hold more exchangeable K and reduces leaching losses of K in soil [55]. In NPK incorporated plants, the nutrients were added externally to the soil, thus there was an increase in soil nutrient level. These results corroborate the findings of previous studies, where the addition of manures and inorganic fertilizers has improved the nutrient status of soil [56].

The variation in micronutrient levels in soil with the organic and inorganic amendments might have been related to changes in the soil environment such as lower soil pH and higher organic matter content. The manures not only supply significant amounts of micronutrients to the soil, but also enhance biological and chemical reactions that result in the dissolution of non-available micronutrients [41]. Moreover, the addition of manures and RSC led to the formation of metal complexes with organic matter that increased the availability of micronutrients by decreasing their adsorption, fixation, and precipitation. The application of N also resulted in lowering the soil pH, which enhanced the availability of DTPA-extractable micronutrients. Similar results have been reported in one of the studies where a significant increase in DTPA-extractable micronutrients has been recorded with the combined application of manures and recommended dose of fertilizers under a rice-wheat system [53].

4.6. Correlation among different soil properties and with yield

Use of organic manure offers the twin benefits of soil quality and fertility enhancement while meeting a part of nutrient needs of crops [57]. Organic manure improves soil structure and water-holding capacity, resulting in more extensive root development, yield and enhanced soil micro-flora and fauna activity, which results in the availability of micronutrients available to plants [58]. Thus, the changes in soil properties observed due to the application of organic manures resulted in an improved correlation between grain and straw yield with soil properties. However, this study does not involve the multi-fertilizers' setup which could be considered as a future prospect of the work.

5. Conclusions

The present study indicates that the partial substitution of inorganic fertilizers with organic manures, especially poultry manure in basmati rice improved plant growth parameters, crop yield, soil properties and nutrient build-up. The addition of poultry manure along with 75% N showed superior results in terms of crop yield and N uptake which indicated the savings of 25% N through the application of organic manures. Similar results were observed for zinc, copper, iron and manganese uptake. The presence of higher N content in poultry manure favoured the vegetative growth as well as improved the nutrient status of the crop. Additionally, the correlation studies showed that the presence of adequate amount of organic carbon in soil correlated well with the nutrient level especially of N, K and Zn in soil. Also, the yield component of basmati exhibited positive correlation with soil P, Cu and Fe which was due to the improved soil properties resulting from the combined use of organic manures along with inorganic fertilizers. The release of nutrients from decomposition of organic manures particularly poultry manure acted as a major factor for these results. Thus, present study inferred that the balanced application of organic (poultry manure) and inorganic (N) fertilizer may prove beneficial for sustainability of basmati rice and to retain the soil fertility. A part from the strengths of the present study, a major limitation is the variation in nutrient levels present in farmyard manure which have different effects on the soil properties, plant growth and yield. Thus, before the application of organic manures, different levels of nutrients should be evaluated properly.

Author contribution statement

Salwinder Singh Dhaliwal, Vivek Sharma, Vibha Verma, Manmeet Kaur - Conceived and designed the experiments, Wrote the paper.

Prabhjot Singh, Manmeet Kaur, Vibha Verma - Performed the experiments.

Salwinder Singh Dhaliwal, Vivek Sharma, Arvind Kumar Shukla, Ahmed Gaber, Akbar Hossain - Analyzed and interpreted the data, Contributed reagents, materials, analysis tools or data.

Funding statement

This work was supported by Taif University, P.O. Box 11099, Taif 21,944, Saudi Arabia.

Data availability statement

Data will be made available on request.

Declaration of interest's statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Additional information

No additional information is available for this paper.

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